Scrambled states survive local Pauli noise up to the hashing entropy

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Scrambling spreads quantum information over high-weight Pauli operators, which local noise damps exponentially, so scrambled encodings are expected to be fragile. We show that they survive up to a sharp threshold. Two pure states drawn independently from any state 2-design, such as random stabilizer states, pass through one layer of i.i.d. Pauli noise. They remain distinguishable until the error entropy reaches one bit per qubit, then become indistinguishable through a Gaussian crossover. For depolarizing noise this threshold, $r = 0.252$, lies far below the $r = 0.423$ certified by purity bounds.

Introduction.—Random quantum states model the output of scrambling dynamics and of generic quantum circuits. A state 2-design, an ensemble whose first two moments match those of Haar-random states, captures their low-order statistics; random stabilizer states are a standard example [1]. Such states are delocalized: in the Pauli basis, almost all of their weight sits on operators that act on a constant fraction of the qubits. Local depolarizing noise of strength r multiplies a weight-w Pauli operator by $(1-r)^w$, so noise removes almost all of the Pauli weight of a scrambled state. This picture underlies the exponential cost of quantum error mitigation (QEM). Takagi, Tajima, and Gu [2] and Tsubouchi, Sagawa, and Yoshioka [3] proved that recovering ideal expectation values requires exponentially many samples in circuit depth. Related cost analyses appear in Refs. [4, 5]. Quek *et al.* [6] showed that random circuits reach this regime at very shallow depth.

Here we ask how much information a scrambled encoding retains after one layer of noise, and we find a sharp answer. Two pure states ψ_1, ψ_2 are drawn independently from a state 2-design, and both pass through the same Pauli channel $\mathcal{N}$. An observer who knows the two candidates measures the output to decide which one was prepared. With equal priors, the optimal success probability is $(1+\overline{T})/2$, where

$$\overline{T} = \mathbb{E}_{\psi_1,\psi_2} \tfrac{1}{2} \|\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2)\|_1. \tag{1}$$

We prove that, at every system size, $\overline{T}$ is bracketed by a single classical quantity: the probability that the Pauli error has surprisal below n. For i.i.d. noise, this places a sharp threshold at the hashing point, one bit of error entropy per qubit, with a universal Gaussian crossover. For local depolarizing noise the threshold is $r_c = 0.252$. Up to this strength the encoded states remain almost perfectly distinguishable, even though the noise has already damped their Pauli weight exponentially.

Two observations place this result in context. First, the threshold exists because no randomization follows the noise. Averaging $U^\dagger \mathcal{N}(U \cdot U^\dagger) U$ over a unitary 2-design turns any Pauli channel into global depolarization $\rho \mapsto \theta\rho + (1-\theta) I/d$, with $\theta = (d^2 p_I - 1)/(d^2-1)$ [7], which decays exponentially in n for every $r > 0$. Without this twirl, the optimal measurement exploits the local structure of the noise. At $n = 8$ and $r = 0.2$, the distance is 0.736, compared with about 0.272 after twirling [Fig. 1(a)]. Second, purity cannot resolve the threshold. The purity argument behind Ref. [6] weighs all errors alike and certifies loss only for $(1-r)^2 < 1/3$, i.e., $r > 0.423$. Ibarrondo and Stilck França [8] and Liu *et al.* [9] obtained the same two values, 0.25 from below and 0.42 from above, and left the interval between them open. Our bound closes this interval. Sommers *et al.* [10] found the same hashing transition, with an $n^{-1/2}$ window, in the coherent information of random codes.

Information-spectrum bracket.—Let $d = 2^n$, $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$, and

$$\mathcal{N}_p(\rho) = \sum_{P \in \mathcal{P}_n} p_P P \rho P, \qquad s(P) = -\log_2 p_P, \tag{2}$$

with p an arbitrary distribution, so errors may be correlated, and with the surprisal s distributed as $P \sim p$. Logarithms are base two. Each $\psi_i = |\psi_i\rangle\langle\psi_i|$ is drawn independently from an exact state 2-design on the full space,

$$\mathbb{E}\psi_i = I/d, \qquad \mathbb{E}\psi_i^{\otimes 2} = (I+F)/[d(d+1)], \tag{3}$$

where F swaps the two copies; the two ensembles may differ.

Theorem 1. *For every Pauli distribution p and every real a,*

$$\mathbb{P}[s < n-a] - 2^{-a} \le \overline{T} \le \mathbb{P}[s < n+a] + 2^{-a/2}/\sqrt{2}. \tag{4}$$

More generally, for any set $\mathcal{G} \subseteq \mathcal{P}_n$ and any partition $\{\mathcal{C}_j\}$ of $\mathcal{P}_n$, with $m_j = \sum_{\mathcal{C}_j} p_P$ and $v_j = \sum_{\mathcal{C}_j} p_P^2$,

$$\overline{T} \ge p(\mathcal{G}) - |\mathcal{G}|/d, \tag{5}$$

$$\overline{T} \le \sum_j \min\left\{ m_j, \sqrt{\frac{d^2 v_j - m_j^2}{2(d+1)}} \right\}. \tag{6}$$

Proof. Fix ψ_1 and let Π project onto the span of $\{P|\psi_1\rangle : P \in \mathcal{G}\}$. For each realization, $\frac{1}{2}\|\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)\|_1 \ge \operatorname{tr}\Pi[\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)]$. The first term is at least $p(\mathcal{G})$. Averaged over the independent ψ_2, the second is $\operatorname{tr}\Pi/d \le$

$|\mathcal{G}|/d$, using only $\mathbb{E}\psi_2 = I/d$. Averaging over ψ_1 gives (5). This is the typical-image argument of Ref. [8], applied to an arbitrary set of Pauli errors.

For the upper bound let $\Delta = \psi_1 - \psi_2$ and $\mathcal{N}_j(\rho) = \sum_{\mathcal{C}_j} p_P P\rho P$. Pauli orthogonality and (3) give

$$\mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\psi_i)^2 = \frac{dv_j + m_j^2}{d+1}, \qquad \mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\psi_1)\mathcal{N}_j(\psi_2) = \frac{m_j^2}{d},$$
$$\mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\Delta)^2 = \frac{2d}{d+1}v_j - \frac{2m_j^2}{d(d+1)}. \tag{7}$$

Jensen's inequality and $\|A\|_1 \le \sqrt{d}\|A\|_2$ bound $\mathbb{E}\frac{1}{2}\|\mathcal{N}_j(\Delta)\|_1$ by the square root in (6), and $\|\Delta\|_1 \le 2$ bounds it by m_j; summing over j gives (6). For (4), take $\mathcal{G} = \{s < n - a\}$, which has fewer than $d2^{-a}$ elements, and split the partition at $s = n + a$, where the tail has $v \le 2^{-n-a}$. □

We have checked this argument in Lean 4 with Mathlib [11, 12]. Taking Eq. (5) and the two-class case of Eq. (6) as hypotheses, the formal proof derives Eq. (4) and the Gaussian estimate of Corollary 1 below. The counting steps, the algebra from Eq. (7) to the square root in Eq. (6), and the examples and thresholds quoted below are proved in full [13].

The theorem has a simple reading. The error images $P|\psi_1\rangle$ of a scrambled state are nearly orthogonal, and a d-dimensional space holds at most d of them. Errors with $p_P > 1/d$, i.e., $s(P) < n$, number fewer than d, so their images fit into the space. The projector onto these images then separates ψ_1 from an independent ψ_2. The remaining errors are too numerous to fit, and their contributions average out. Distinguishability therefore equals the probability of a likely error, up to a surprisal window of width a and exponentially small corrections.

Taking the trivial partition, $\mathcal{C} = \mathcal{P}_n$, reduces the upper bound (6) to the purity of the noisy output, $\overline{T} \le [d\sum_P p_P^2/2]^{1/2}$. This is the quantity that controls the QEM bounds of Ref. [6]. For local depolarizing noise, $\sum_P p_P^2 = [(1+3(1-r)^2)/4]^n$, which falls below d^{-1} only for $r > 1 - 1/\sqrt{3} = 0.423$. The purity bound is loose below this value because it places the likely errors and the unlikely ones in the same second moment. Separating the two classes at surprisal n removes this looseness and lowers the threshold to the hashing point.

Sharp threshold.—For i.i.d. noise, $p = \boldsymbol{p}^{\otimes n}$ with a single-qubit distribution $\boldsymbol{p} = (p_0, p_x, p_y, p_z)$. The surprisal is then a sum of n independent terms. Its mean is nH, where $H = \mathbb{E}[-\log_2 p_i]$ is the entropy of $\boldsymbol{p}$, and its variance is $n\sigma^2$, where $\sigma^2 = \mathrm{Var}[-\log_2 p_i]$. Taking $a = \epsilon n$ in (4) and applying a large-deviation bound [14] gives

$$\overline{T} = \begin{cases} 1 - e^{-\Omega(n)}, & H < 1, \\ e^{-\Omega(n)}, & H > 1. \end{cases} \tag{8}$$

The threshold $H = 1$ is the hashing point, at which the coherent information $1 - H$ of the Pauli channel with maximally mixed input vanishes [15, 16]. For depolarizing noise $\mathcal{N}_r(\rho) = (1-r)\rho + rI/2$, with $q = 3r/4$ and $\boldsymbol{p} = (1-q, q/3, q/3, q/3)$, the threshold lies at $q_c = 0.189290\ldots$, or $r_c = 0.252386\ldots$. Degenerate codes can transmit quantum information slightly beyond the hashing point [17]; r_c is the threshold for distinguishing two encoded states.

The bracket also resolves the shape of the transition. Let the single-qubit distribution $\boldsymbol{p}_n$ depend on n, with i.i.d. errors at each n, and define $x_n = \sqrt{n}(1 - H_n)/\sigma_n$.

Corollary 1. *If $\boldsymbol{p}_n \to \boldsymbol{p}_*$ with $H(\boldsymbol{p}_*) = 1$ and $\sigma(\boldsymbol{p}_*) > 0$, then for every pair of independent exact 2-design ensembles,*

$$\overline{T}_n = \Phi(x_n) + O\left(\frac{\log n}{\sqrt{n}}\right), \tag{9}$$

where Φ is the standard normal distribution function.

Proof. The Berry–Esseen theorem [18] gives $\mathbb{P}[s < n \mp a] = \Phi(x_n \mp a/\sqrt{n}\sigma_n) + O(n^{-1/2})$, uniformly because the third moment of the surprisal is bounded on a four-letter alphabet and σ_n is bounded away from zero. Insert into (4) with $a = 2\log_2 n$ and use the bounded density of Φ. □

Near threshold, distinguishability therefore depends on r, to leading order, only through $\sqrt{n}(r - r_c)$, and it follows the universal profile Φ. Figure 1 shows the finite-size bracket at $n = 8$, exact Haar averages, and the approach to Φ at large n.

Above threshold, the distance decays exponentially. Grouping errors into empirical types in (6) gives $\overline{T}_n \le \mathrm{poly}(n)2^{-nE(\boldsymbol{p})}$, with [14]

$$E(\boldsymbol{p}) = \min_t \{D(t\|\boldsymbol{p}) + \tfrac{1}{2}[H(t) - 1]_+\}$$
$$= \max_{0\le s\le 1} \frac{s}{1+s}[H_{1+s}(\boldsymbol{p}) - 1], \tag{10}$$

where D is the relative entropy and H_α is the Rényi entropy. This equals the universal$_2$ privacy-amplification exponent [19]. It also follows, without the polynomial prefactor, from decoupling without smoothing [20], because the family $X \mapsto \sum_P X_{PP} P\psi P$ is 1-randomizing under (3). Decoupling [9, 20] therefore also yields decay for $H > 1$, whereas the bracket supplies the location of the transition and its Gaussian profile. Near threshold, the lower bound (5), applied to types with $D(t\|\boldsymbol{p}) + H(t) < 1$, matches E to leading order. Let R_+ and R_- denote the lim sup and lim inf of $-n^{-1}\log_2 \overline{T}_n$. Both satisfy

$$R_\pm = \kappa(r - r_c)^2 + O((r - r_c)^3),$$
$$\kappa = \frac{9}{32\ln 2\, q_c(1 - q_c)} = 2.64\ldots \tag{11}$$

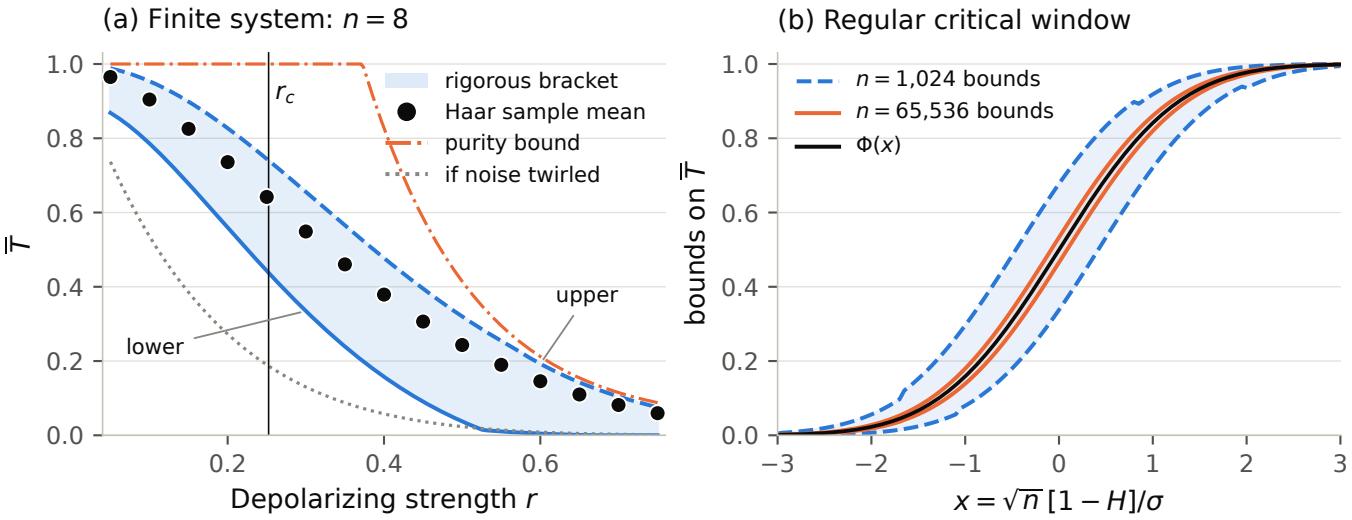


FIG. 1. Distinguishability after one layer of local depolarizing noise. (a) At $n = 8$, the shaded band lies between the projector lower bound (solid) and the two-class upper bound (dashed) of Eqs. (5) and (6), optimized over error-weight cutoffs; the thin vertical line marks r_c. Points are averages over 64 independent Haar-state pairs from full density matrices, with one-standard-error bars; the same pairs are used at every r. The dash-dotted curve is the purity bound, which drops below one only for $r \gtrsim 0.37$ at this size and certifies decay only above 0.423 asymptotically. The dotted gray curve is the depolarizing parameter θ reached if a further 2-design unitary twirled the noise; it equals the twirled distance up to a factor $1 - O(1/d)$. The lower bound is zero for $r \geq 2/3$, where no single error has probability above $1/d$. (b) The bracket (4) with $a = 2\log_2 n$, evaluated from binomial distributions at fixed $x = \sqrt{n}(1 - H)/\sigma$ for $n = 1024$ and 65536, approaches $\Phi(x)$ (black).

Correlated noise.—The bracket applies to any Pauli distribution. For correlated errors, the full information spectrum, rather than the entropy, determines the answer. Global depolarization $\mathcal{N}(\rho) = \theta\rho + (1 - \theta)I/d$ is a Pauli channel with $p_I = \theta + (1-\theta)/d^2$ and $p_P = (1-\theta)/d^2$ otherwise. Its entropy rate approaches $2(1 - \theta)$, which exceeds one for $\theta < 1/2$. Nevertheless, $\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2) = \theta(\psi_1 - \psi_2)$, so $\overline{T} \to \theta$. The bracket recovers this limit. With $a = n/2 - 1$, only the identity lies below surprisal $n - a$, all other errors lie above $n + a$, and $|\overline{T} - \theta| \leq 2^{-n/4+1}$.

Design dependence.—The bracket uses only the first two moments of the ensemble, so the threshold and the Gaussian profile are identical for every 2-design. Two finer features lie beyond the bracket: the limit at a degenerate critical point, which depends on the ensemble, and the decay rate away from threshold.

Consider first a degenerate critical point. Complete dephasing, $\boldsymbol{p}_* = (1/2, 0, 0, 1/2)$, has $H = 1$ but $\sigma = 0$, so Corollary 1 does not apply. The uniform union of a complete set of $d + 1$ mutually unbiased bases (MUBs) is an exact 2-design [21]. Take the computational basis as one of them. Its d states survive dephasing, and all others map to I/d, so

$$\overline{T}_{\mathrm{MUB}} = \frac{(d-1)(2d+1)}{d(d+1)^2} \to 0, \qquad \overline{T}_{\mathrm{Haar}} = \frac{d-1}{2d-1} \to \frac{1}{2}. \tag{12}$$

For Haar states the dephased outputs are diagonal, so their trace distance is the total-variation distance between two independent flat Dirichlet vectors. For single coordinates $X, Y \sim \mathrm{Beta}(1, d-1)$, $\mathbb{E}\min(X, Y) = (2d-1)^{-1}$. Two ensembles with identical second moments thus reach different limits (Fig. 2). This refines the observation of Ref. [8] that dephasing leaves computational-basis states uncontracted. The variance condition must also hold in the limit. Take $\epsilon_n = n^{-6}$ and $\boldsymbol{p}_n = (1/2 - \epsilon_n + \delta_n, \epsilon_n, \epsilon_n, 1/2 - \epsilon_n - \delta_n)$, where $\delta_n = O(\sqrt{\epsilon_n \log(1/\epsilon_n)})$ solves $H(\boldsymbol{p}_n) = 1$. The n-qubit channel then differs from complete dephasing by at most $n\|\boldsymbol{p}_n - \boldsymbol{p}_*\|_1 \to 0$ in diamond norm. MUB inputs therefore still give $\overline{T}_n \to 0$, although $\sigma_n > 0$ for every n.

Consider next the decay rate away from threshold. A random stabilizer state is $\psi = d^{-1}[I + \sum_{P \neq I} z_P P]$, with $z_P \in \{0, \pm 1\}$ and $\mathbb{E}|z_P| = 1/(d+1)$. Local depolarization multiplies the weight-w terms by $(1 - r)^w$, so

$$\overline{T}_n^{\mathrm{stab}} \leq \frac{[1 + 3(1-r)]^n - 1}{2^n + 1}. \tag{13}$$

At $r = 0.9$, this distance decays at a rate of at least $1 - \log_2(1.3) = 0.621$, faster than the rate $E = 0.479$ guaranteed by Eq. (10). Random stabilizer states form a 3-design [1], so agreement with Haar states in three moments does not make Eq. (10) tight. Whether the Haar rate also exceeds E away from threshold remains open.

Discussion.—Scrambling does not make encoded information fragile to local Pauli noise. Two scrambled states remain distinguishable up to the hashing point, at which the error distribution carries one bit per qubit,

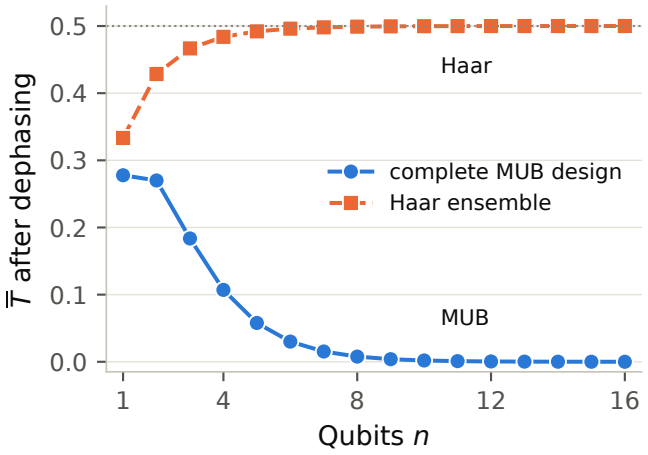


FIG. 2. Design dependence at a degenerate critical point. Under complete dephasing both ensembles are exact 2-designs and the error entropy is one bit per qubit, yet Eq. (12) gives different limits. The dotted gray line marks 1/2.

and the transition has a universal Gaussian shape. A single bracket, built from the error distribution and the Hilbert-space dimension alone, establishes this for every 2-design and also covers correlated noise.

This result separates two regimes of noisy quantum computation. QEM costs grow exponentially at large depth for any noise strength [2, 3, 6], and the purity estimates behind those bounds weigh all errors alike. After a single noisy layer, however, the optimal measurement exploits the likely errors, which purity estimates miss, and the threshold moves from 0.423 to 0.252. Splitting the error distribution at surprisal n may therefore sharpen other purity-based bounds. The decisive open question is how the threshold evolves in deeper circuits, where later scrambling layers partly twirl earlier noise toward global depolarization. Tracking the information spectrum layer by layer would determine whether a finite noise threshold survives at depth, or whether it is erased as the twirling argument suggests.

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Data availability.—The data that support the findings of this article, the code that generates every figure and numerical check, and the Lean formalization accompany this submission and will be deposited in a public repository [13].

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